AI Quiz System (SmartLearn AI)

System Architecture Specification & Technical Documentation — Category: AI / ML & EdTech Platform

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Repository: https://github.com/rakshapatil526/AI-Quiz-System

Status: Verified Repository

Doc ID: SPEC-AI-QUIZ-SYSTEM

Date: March 2026

1. Problem Statement & Research Objectives

Students face immense cognitive load when digesting voluminous textbooks, research papers, and lecture slides. Traditional study tools rely on pre-existing, static question banks that fail to align with a student's specific syllabus. SmartLearn AI solves this by ingesting uncurated study materials and automatically generating structured study notes, multi-perspective summaries, interactive flashcards, RAG-grounded study chatbots, and dynamic quizzes with no dependency on static banks.

2. Multi-Stage Document Ingestion & NLP Pipeline

```
[Student Document (PDF / DOCX / PPTX / TXT)]
       ■
       ■■■ PyMuPDF (`fitz`): Direct text & layout extraction
       ■■■ Scanned Page Detector: Triggers Tesseract OCR via pytesseract (<35 chars/page)
       ■■■ Semantic Text Chunking: Overlapping passages tagged with source page numbers
       ■
       ▼
[NLP Linguistic Feature Extraction Engine]
       ■■■ Tokenization, Stop-word elimination, N-Gram analysis
       ■■■ Definition Identification: Pattern-matched conceptual terms
       ■■■ Formulas & Facts Extractor: Regex capture of complexity notation & equations
       ■
       ▼
[Generative Assessment Pipeline]
       ■■■ MCQs (4 options: 1 correct + 3 grounded distractors)
       ■■■ True/False, Fill in the Blanks, and Short Answer Questions
       ■■■ 4 Difficulty Tiers: Easy, Medium, Hard, and Expert
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3. Machine Learning Analytics (Scikit-Learn)

ML Subsystem	Algorithm & Implementation	Features & Performance Results
Performance Predictor	RandomForestClassifier (joblib serialized)	Features: topic_accuracy, overall_accuracy, difficulty, attempt_num. Accuracy: 72%, Recall: 89%, F1: 0.816
Learner Clustering	K-Means Clustering (k=4)	Personas: Analytical Master, Speedy Skimmer, Deep Contemplator, Foundational Explorer
RAG Document Chatbot	TF-IDF + Cosine Vector Matching	Retrieves top-k context passages from document chunks with source page citations

4. Technology Stack & Architecture

- **Backend Engine:** Python 3, Django, Django REST Framework, Scikit-learn, PyMuPDF, pytesseract, ReportLab.
- **Frontend Client:** React 18, Vite, Tailwind CSS, Lucide Icons, interactive quiz timers, confetti animations.
- **Evaluation:** Model evaluated on train/test split with full confusion matrix and precision/recall trade-off analysis.